

"Math Is Cool" Championships — 2025-26

5th Grade — Feb/March 2026

GENERAL INSTRUCTIONS applying to all tests:

- *Good sportsmanship is expected throughout the competition by all involved (competitors and observers). Display of poor sportsmanship will result in disqualification.*
- *Competitors may not use calculators or any other aids on any portion of this contest.*
- *Unless stated otherwise:*
 - *All answers are integers, and any non-integer answers will be "coded" as integers.*
 - *For 5th grade and up, all fractions and ratios must be reduced to simplest form, all radicals must be simplified, and all denominators must be rationalized.*
 - *Do not round or approximate answers. Leave answers in terms of π or other irrational quantities (e.g., $\sqrt{2}$), where applicable.*
- *Units are not necessary as part of your answer, However, if you choose to use units, they must be correct.*
- *Record all answers on the colored cover sheets in the answer column only.*
- ***Be sure that the student name, school, team number, etc. has been filled out at the top of each answer sheet.***
- *Tests will be scored as a 0 if answers are not recorded correctly on the answer sheets.*
- *Blank answer sheets and answer sheets with no name will be scored as a 0.*

FINAL SCORES AND AWARDS

Individual awards are determined by both the Mental Math and Individual Test scores. Individual ties are broken based on the following, in this order: total scaled individual points, total number of correct answers on the Individual Test, Mental Math raw score, number of correct answers from Individual Test #31-40, number of correct answers from Individual Test #16-30, highest numbered question answered correctly on the Individual Test working backwards from #40.

Team (School) awards are based on the highest score from amongst each of the school's "teams of 4 students" in each event and is calculated as $2 \cdot (\text{Sum of highest 3 Mental Math scores}) + 2 \cdot (\text{Multiple Choice}) + 6 \cdot (\text{Team}) + 1 \cdot (\text{Triple Jump}) + 1 \cdot (\text{College Bowl})$, for approximate weights of 25%, 20%, 30%, 15% and 10% respectively. Team ties are broken based on highest event score in order of the events, starting with Mental Math.

MENTAL MATH TEST - 30 sec./quest., 8 problems, ~8%/25% of individ./team scores

The proctor will read each question twice. You may not do any writing or talking while arriving at a solution. Record only your answer on your answer sheet. You may not change, cross out, erase, or write over an answer once you have written it down. The maximum wait time is 30 seconds after completion of the second reading of the question. Correct answers receive 1 point.

INDIVIDUAL TEST - 35 minutes, 40 problems, ~92% of individual score

When you are prompted to begin, tear off the colored answer sheet and begin testing. No talking during this individual test. You will be given a 5 minute time warning. Correct answers receive 2 points for problems 1-30 and 3 points for 31-40 (in the scaled score).

"Math Is Cool" Championships — 2025-26

5th Grade — Feb/March 2026

Final Score (out of 8)

Room #

School Name

Student Name

Team #

Mental Math - ~25% of team score & ~8% of individual score

All students in the room will concurrently be asked the same eight questions in this individual test. When it is time to begin, the proctor will read the first question twice. You may not do any writing or talking while arriving at a solution. Once you have a solution, record it on the sheet in front of you. **You may not change or cross out answers once you have written an answer down. If there are eraser marks, write-overs, or crossed-out answers, they will be marked wrong.** Once all students have laid their pencils on the desk, another question will be asked. If a student doesn't lay his or her pencil down, the maximum wait time is 30 seconds after completion of the second reading of the question before the next question is read. You may continue to work on a problem (in your head) while the next question is being read. The raw score is 1 point per correct answer.

STUDENT: DO NOT WRITE IN SHADED REGIONS (or anywhere else, other than the answer box)

Answer		Scorer 2	Scorer 1
		0 or 1	0 or 1
1			
2			
3			
4			
5			
6			
7			
8			
5 th Grade	TOTAL:		

"Math Is Cool" Championships — 2025-26

5th Grade — Feb/March 2026

Key

Mental Math Contest – Answer Key

30 seconds per question – ~25% of team score & ~8% of individual score

SCORERS — Write-overs, Cross-outs, and Erasures Must be Marked Incorrect (0)
Bracketed items [...] in the answer key are optional.

5th Grade

Answer		
1	76	What is the sum of twenty-five and fifty-one?
2	24 [inches]	In inches, what is the perimeter of a square with a side length of six inches?
3	34	What is fifty percent of sixty-eight?
4	55 [= median]	Charlie scores a fifty, forty-eight, fifty-five, sixty and sixty-two on his most recent math tests. Which score is the median?
5	20 [= x]	X plus fourteen equals thirty-four. What is x?
6	1 [feet, or foot]	A right triangle has a base of four feet and a height of five feet. A rectangle with a base of ten feet has the same area as the triangle. What is the height of the rectangle in feet?
7	42 [= total value]	The letters of the alphabet A through Z are each assigned an integer from one to twenty-six in order. What is the total value of the letters in the word MATH, spelled M-A-T-H, when added together?
8	224 [= 8th term]	The first two terms of a sequence are two and five. Each of the following terms in the sequence is equal to the sum of all the terms that come before it in the sequence. What is the eighth term in the sequence?

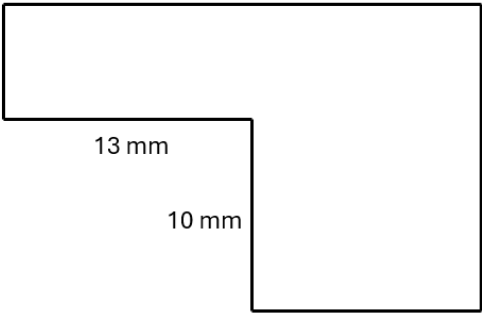
"Math Is Cool" Championships — 2025-26

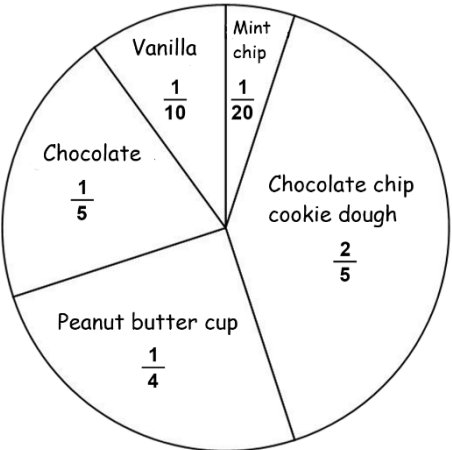
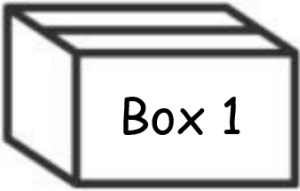
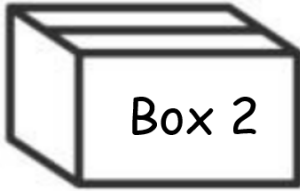
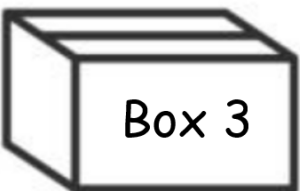
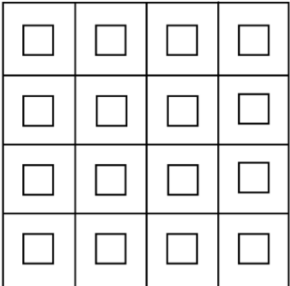
5th Grade — Feb/March 2026

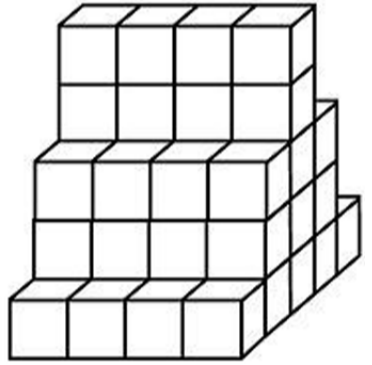
Individual Contest

Record all answers on the colored cover sheet. 35 minutes, 40 problems, ~92% of individual score.

No talking during this individual test. A 5-minute time warning will be given.

	Questions 1-30: 2 points each
1	What is 452 multiplied by 8?
2	If $x - 15 = 16$, what is the value of x ?
3	What is the place value of the digit 5 in the number 358?
4	As an integer, what is the value of the fraction? $\frac{27}{9}$
5	A triangle has angle measures of 40° and 60° . In degrees, what is the measure of the third angle?
6	Delisha has 8 apples and 2 oranges. How many apples are there for every orange?
7	What is 10.58 rounded to the nearest whole number?
8	A square has a side length of 3 centimeters. In centimeters, what is the perimeter of the square?
9	What is the mode of the data set: {40, 45, 46, 46, 50, 52}?
10	Ginger has 3 boxes of crayons. Each box has 6 crayons inside. How many crayons does she have in total?
11	Sophie has some marbles. Her friend gives her 5 more, and now she has 12. How many marbles did Sophie have to start with?
12	Which of the following is a factor of 32? 3, 6, 8, 10, 12, 17
13	How many inches are there in 4.5 feet?
14	In square millimeters, what is the area of the figure shown?
Continued on next page.	

15	Find the missing number in the sequence: 11, 55, _____, 1375, 6875, ...
16	The following graph shows the favorite ice cream flavors of Mrs. Schuh's 5th grade students. What percentage of students have either chocolate or chocolate chip cookie dough as their favorite flavor? 
17	If a movie starts at 2:05 p.m. and ends at 4:47 p.m. the same afternoon, how many minutes did the movie last?
18	Each of the boxes has one true statement and one false statement written below them. Which box number contains the prize?  1. The prize is not in here. 2. The prize is not in Box 3.  1. The prize is in Box 2. 2. The prize is not in Box 1.  1. The prize is not in here. 2. The prize is not in Box 1.
19	You have 3 ten-dollar bills, 4 five-dollar bills, and 2 one-dollar bills. How many dollars do you have in total?
20	What is the greatest common factor of 48 and 60?
21	A 5th grade class is organizing a fundraiser. They decide to sell snack packs to raise money for a field trip. Each snack pack contains the following: 2 juice boxes, 1 granola bar, and 3 fruit snacks. If the class has 120 juice boxes, 60 granola bars, and 150 fruit snacks, what is the maximum number of complete snack packs the students can make with the supplies they have?
22	What is the positive difference between the largest and the smallest prime factors of 220?
23	How many squares of any size are in the following figure? 
24	Hyrum has 72 books that he wants to organize evenly on 6 shelves. After he is done organizing the books, he has 12 books left over that he will donate. How many books did he put on each of the 6 shelves?
Continued on next page.	

25	Brooklyn recorded the number of goals in each of her 5 soccer games: 2, 3, 4, 5, 6. How many goals does she need to score in her 6 th game to have an average of 5 goals per game overall?
26	The following figure is made up of 52 cubes of edge length 2 inches, which have been glued together into a symmetric shape. In square inches, what is the total surface area of the figure? 
27	A cookie recipe requires $3\frac{1}{2}$ cups of flour for every $1\frac{1}{2}$ cups of sugar. Harlen wants to triple the recipe to make enough for a class party. How many total cups of sugar and flour combined will be needed?
28	What is the 20 th number in this pattern? 2, 5, 8, 11, ...
29	An integer from 1 to 50 (including 1 and 50) is randomly selected. What is the probability as a percentage that it is a multiple of 5 or 7?
30	An animal shelter has cats and dogs. There are four more cats than dogs in the shelter. If there are a total of 28 animals, how many of them are cats?
Challenge Questions: 3 points each	
31	In the following pattern of dates, what would complete the last date? Enter your answer as a 2-digit integer, where the two digits would fill in the two blanks. January 2, 2021 February 8, 2082 March 9, 2093 April 1, 2014 May 4, 20__
32	Forty cards have been labeled with the integers 1 through 40. Four different cards are randomly selected, and placed in order of their selection from left to right. The probability that the number on each card in order from left to right is greater than the previous number can be written as a simplified fraction A/B . What is $A + B$?
33	An iron bar is 2.9 meters long. If it is cut into pieces that are 3 centimeters long, how many whole pieces can be made?
34	The digits 2 and 4 can be used to create 8 distinct three-digit positive integers, using each digit (2 or 4) from 0 to 3 times. What is the mean of these 8 integers?
35	In a geometric sequence of positive numbers, if the 21 st term is 500 and the 39 th term is 32, what is the 33 rd term?
36	How many positive three-digit integers contain the digit 9 at least once?
Continued on next page.	

"Math Is Cool" Championships — 2025-26

KEY

Individual Contest – Answer Key

SCORERS: Bracketed [...] items in answer key are optional. Just mark the score as 0 or 1 and add up those values to reflect total correct.
First Scorer – use the right-hand columns so 2nd scorer can do a blind scoring.

	Answer
1	3616
2	31 [=x]
3	50
4	3
5	80 [degrees]
6	4 [apples]
7	11
8	12 [cm]
9	46 [= mode]
10	18 [crayons]
11	7 [marbles]
12	8 [= a factor]
13	54 [inches]
14	270 [square mm, mm ²]
15	275

	Answer
16	60 [%]
17	162 [minutes]
18	3 [= box number]
19	52 [dollars]
20	12 [= GCF]
21	50 [snack packs]
22	9 [= positive difference]
23	46 [squares]
24	10 [books]
25	10 [goals]
26	424 [sq inches]
27	15 [cups]
28	59 [= 20 th number]
29	32[%]
30	16 [cats]

	Answer
31	45
32	25 [= A + B]
33	96 [whole pieces]
34	333 [= mean]
35	80 [= 33 rd term]
36	252 [3-digit integers]
37	324 [= A]
38	11 [= sum]
39	175 [nickels]
40	1014

5th Grade —
Feb/March 2026

"Math Is Cool" Championships — 2025-26

Total Correct (all columns)

Room #

SCHOOL NAME

STUDENT NAME

Team #

Individual Contest - Score Sheet

STUDENTS: DO NOT WRITE IN SHADED REGIONS

	Answer	1 or 0	1 or 0
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
1-15 TOTAL:			

	Answer	1 or 0	1 or 0
16			
17			
18			
19			
20			
21			
22			
23			
24			
25			
26			
27			
28			
29			
30			
16-30 TOTAL:			

	Answer	1 or 0	1 or 0
31			
32			
33			
34			
35			
36			
37			
38			
39			
40			
31-40 TOTAL:			

5th Grade — Feb/March
2026

Scorers: Just score as 0 or 1 and add up those values (i.e., just work with number correct).

"Math Is Cool" Championships — 2025-26

5th Grade — Feb/March 2026

Team Multiple Choice Contest

USE THE FOLLOWING INFORMATION TO SOLVE PROBLEMS #1 THROUGH #2.

Mrs. Tamayo has her 5th grade students keep track of how many minutes of free reading they do each day. The following table shows a sample of four of her students for the week of January 12th, 2026. All data are given in minutes.

	Monday	Tuesday	Wednesday	Thursday	Friday
Abinaya	24	30	18	22	?
Jace	32	28	25	30	35
Xizhen	18	20	22	24	36
Neev	40	36	30	28	26

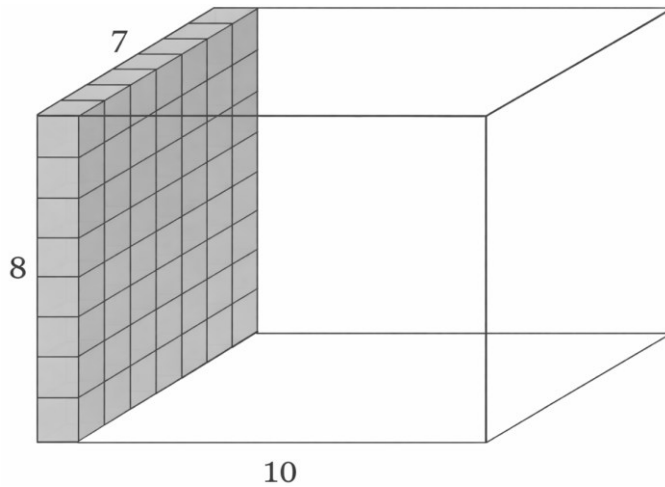
- 1 Abinaya and Xizhen read the same number of minutes for the total week. How many minutes did Abinaya read on Friday?
- A) 24 B) 25 C) 26 D) 28 E) None of the above.

- 2 What was the mean number of minutes that the students read for the total week?
- A) 120 B) 135 C) 137.5 D) 145 E) None of the above.

Continued on Next Page

USE THE FOLLOWING INFORMATION TO SOLVE PROBLEMS #3 THROUGH #6.

The following rectangular prism has side lengths of 7, 8 and 10 units, as shown. The prism is being filled with unit cubes.

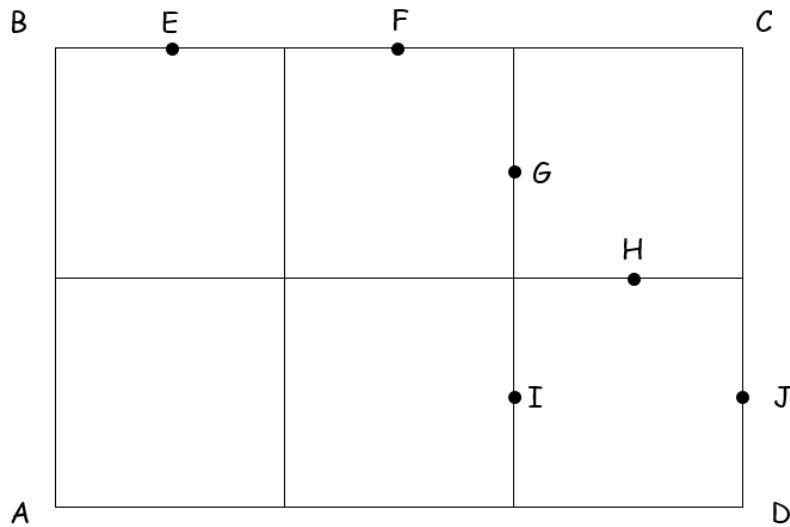


3	How many unit cubes are currently in the prism? A) 56 B) 80 C) 87 D) 100 E) None of the above.
4	How many more unit cubes need to be added in order to fill the prism completely? A) 56 B) 240 C) 504 D) 560 E) None of the above.
5	Once the entire prism is full of unit cubes, all 6 outer surfaces of the prism will be painted. How many unpainted cubes will there be in the interior of the prism? A) 148 B) 240 C) 248 D) 316 E) None of the above.
6	If all of the cubes with one or more sides painted are put into a bag, and one of them is randomly selected and rolled (like a die), what is the probability that a painted side will be showing on the top after the roll? A) $\frac{1}{3}$ B) $\frac{1}{2}$ C) $\frac{421}{1280}$ D) $\frac{103}{320}$ E) None of the above.

Continued on Next Page

USE THE FOLLOWING INFORMATION TO SOLVE PROBLEMS #7 THROUGH #10.

Rectangle ABCD is made up of six congruent (equal) squares. The side length of each square is one unit. The points E, F, G, H, I and J are located at midpoints of the sides of the squares.



- 7 In units, what is the combined length of all the 1-unit segments that are shown in the figure?
A) 15 B) 16 C) 17 D) 18 E) None of the above.
- 8 In units, what is the shortest distance for an ant to crawl from vertex A to vertex C, if the ant must travel along the edges of the squares only.
A) 4 B) 6 C) 8 D) 9 E) None of the above.
- 9 How many different line segments can be drawn from vertex B to one of the other vertices (A, C, D) or to one of the labeled midpoints (E, F, G, H, I, J), as long as the line segment does not pass through another labeled midpoint or vertex?
A) 7 B) 8 C) 9 D) 10 E) None of the above.
- 10 A thrown dart lands randomly anywhere inside rectangle ABCD. What is the probability that it lands in triangle ACH?
A) $1/6$ B) $2/25$ C) 10% D) 12% E) None of the above.

"Math Is Cool" Championships — 2025-26

5th Grade — Feb/March 2026

Key

Team Multiple Choice Contest – Answer Key

5th Grade

Correct responses are worth 2 points, incorrect responses are worth -1 point, and absence of a response is worth 0 points.

Answer	
1	C
2	C
3	A
4	C
5	B
6	D
7	C
8	E (5)
9	A
10	A

"Math Is Cool" Championships — 2025-26

5th Grade — Feb/March 2026

Final Score (out of 20)

Room #

School Name

Team #

Team Multiple Choice Contest - 15 minutes - ~20% of team score

You will have 15 minutes to answer 10 multiple choice questions as a team. This test is the only test where you will be penalized for incorrect responses. You will receive two points for a correct letter response, zero points for leaving it blank, and minus one point for an incorrect response. When you are prompted to begin, tear off the colored answer sheet, pass out a copy of the test to each team member, and begin testing. **ONLY a letter response should be listed as an answer on this answer sheet. Correct responses are worth 2 points, incorrect responses are worth -1 point, and absence of a response is worth 0 points.**

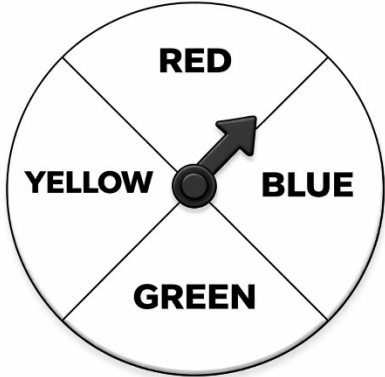
STUDENTS: DO NOT WRITE IN SHADED REGIONS

		Scorer 2	Scorer 1
Answer		-1, 0, or 2	-1, 0, or 2
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
5 th Grade		TOTAL:	

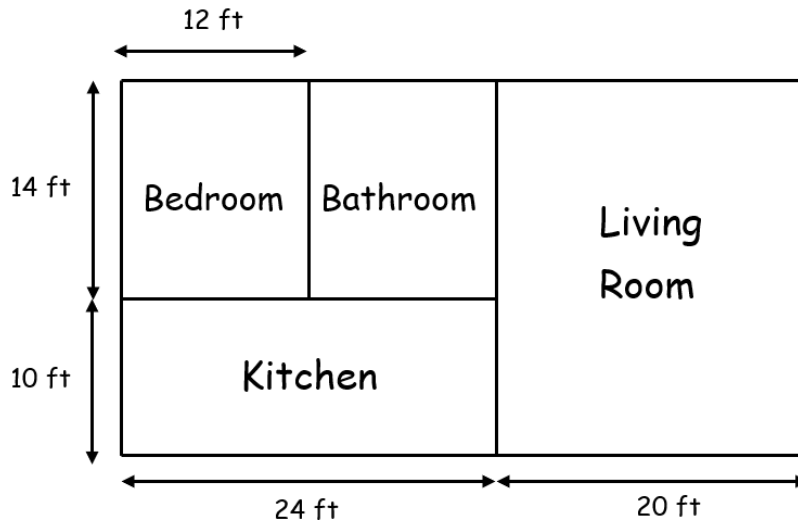
"Math Is Cool" Championships — 2025-26

5th Grade — Feb/March 2026

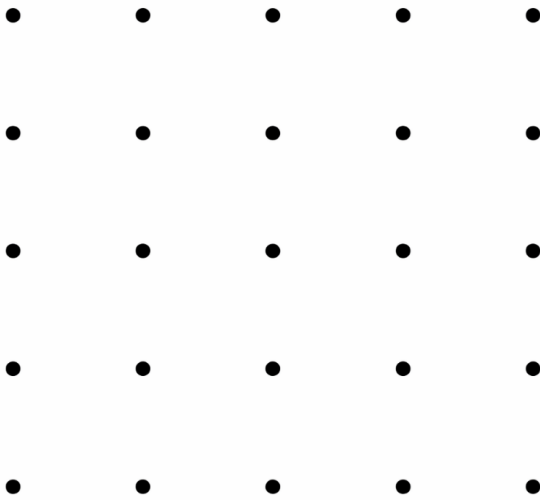
Team Contest

1	A Labubu toy factory makes 84 Labubus in 1 hour. If they work 7 hours a day, how many Labubus do they make in 5 days?
2	What is the sum of the prime numbers between 1 and 15?
3	A square garden has a perimeter of 64 feet. In square feet, what is the area of the garden?
4	A circular spinner is divided into 4 equal sections: red, blue, green, and yellow. When it is spun 100 times, what is the expected number of times the spinner lands on green? 
5	Evaluate: $5(4 + 2) - 3(1 + 2)$
6	A tank can hold 180 gallons of water. It is $\frac{2}{3}$ full. After some water is drained, it is $\frac{1}{4}$ full. How many gallons were drained?
7	The following pyramid is an example of a "self-descriptive" sequence, where each row describes the digits that are in the previous row. For example, the 2nd row tells us that there was one '2' in the first row. The 3rd row tells us that there was one '1' and one '2' in the second row. A total of 6 rows of the pyramid are included. What is the sum of the digits in the 7th row? <pre> 2 1 2 1 1 1 2 3 1 1 2 1 3 2 1 1 2 1 1 1 3 1 2 2 1 1 2 </pre>
Continued on next page.	

- 8 Given the following floor plan of Abigail's new apartment, the area of the bathroom is what percentage of the area of the living room?



- 9 Given the 5×5 array of dots, in how many different ways can four dots be selected to form the vertices of a square?



- 10 Two bricklayers are building a wall. Sarah would take 9 hours to build the wall alone, and Murphy would take 10 hours to build it alone. Working together, they talk too much, which causes their combined output to be reduced by 10 bricks per hour. Together, they build the wall in 5 hours. How many bricks are in the wall?

"Math Is Cool" Championships — 2025-26

5th Grade — Feb/March 2026

Key

Team Contest - Answer Key

5th Grade

Answer	
1	2,940 [Labubus]
2	41 [= sum]
3	256 [square feet]
4	25 [times]
5	21
6	75 [gallons]
7	20 [= sum]
8	35 [%]
9	50 [ways] [squares]
10	900 [bricks]

"Math Is Cool" Championships — 2025-26

5th Grade — Feb/March 2026

Final Score (out of 10)

Room #

School Name

Team #

Team Contest - 15 minutes - ~30% of team score

When you are prompted to begin, tear off the colored answer sheet and give a copy of the test to each of your team members and begin testing. Each problem is scored as a 1 or 0. Record all answers on this colored answer sheet.

STUDENTS: DO NOT WRITE IN SHADED REGIONS

Answer		Scorer 2	Scorer 1
		0 or 1	0 or 1
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
5 th Grade		TOTAL:	

"Math Is Cool" Championships — 2025-26

5th Grade — Feb/March 2026

Linda Moore Triple Jump

1	Natalie has 2 dozen cookies. She sells each cookie for \$2.50. In dollars, how much money will she make if she sells every cookie?
2	In centimeters, what is the diameter of a circle that has a radius of 3.5 centimeters?
3	A bag has 5 red marbles, 4 blue marbles, and 3 green marbles. One marble is chosen at random. What is the probability, expressed as a percentage, that it is not green?
4	A three-digit positive integer has digits that add to 15. The hundreds digit is twice the ones digit. The tens digit equals the hundreds digit. What is the number?
5	A staircase is built with blocks so that step 1 has 1 block, step 2 has 3 blocks, step 3 has 5 blocks, and so on. If the staircase has 12 steps, how many blocks are needed in total?
6	Mr. Savage has 4 markers for every 9 pencils in his classroom. If there are 45 more pencils than markers, how many markers are there?
7	A chocolate candy bar has dimensions of 6 by 8 units. What is the minimum number of breaks, either horizontal or vertical, that are needed to separate it into pieces of 1 by 1 units? Pieces cannot be stacked on top of each other before breaking.
8	The base of an isosceles triangle is 24 inches, and its area is 60 square inches. What is the length in inches of one of the congruent sides?
9	How many subsets of the set of numbers {1, 2, 3, 4, 5, 6, 7, 8} contain both 1 and 2, but do not contain 7?
Continued on next page.	

10 From a standard deck of playing cards, all 13 hearts have been lined up in a row, with positions 1 through 13, as indicated below. Face cards are defined as Jack, Queen and King. The Ace is the high card, so the natural order of the cards from left to right is 2 through 10, Jack, Queen, King and Ace. Assume a value of 15 for the ace and face cards. Otherwise, assume their numerical value.

Given the following clues, determine the order of the 13 cards.

What is the sum of the card values in positions 5, 7 and 8?

Clues:

- No two cards are in their natural order from left to right.
- No face cards are at either end.
- Two face cards are adjacent.
- The King and Ace are separated by one card.
- The 2nd and 3rd cards have a sum of 13.
- The 10th and 11th cards have a sum of 10.
- The 8 is two places to the left of the ace.
- The Jack is two places to the right of the 2.
- The 4 is two places to the right of the 3.
- The 6 is three places to the left of the ace.
- The 9 is somewhere to the right of the 4.
- The 10 is two places to the right of the 5.
- The third card from the right and the third card from the left are both odd.

1st 2nd 3rd 4th 5th 6th 7th 8th 9th 10th 11th 12th 13th

"Math Is Cool" Championships — 2025-26

5th Grade — Feb/March 2026

Key

Linda Moore Triple Jump - Answer Key

5th Grade

Answer	
1	60 [dollars]
2	7 [centimeters]
3	75 [percent]
4	663 [= the no.]
5	144 [blocks]
6	36 [markers]
7	47 [breaks]
8	13 [inches]
9	32 [subsets]
10	27 [= sum]

"Math Is Cool" Championships — 2025-26

5th Grade — Feb/March 2026

Final Score (out of 10)

Room #

School Name

Team #

Linda Moore Triple Jump - 15 minutes - ~15% of team score

When you are prompted to begin, tear off the three colored answer sheets and give a copy of the test to each of your team members and begin testing. Record all answers on this colored answer sheet. This Submittal #1 will be collected after 5 minutes.

SUBMITTAL #1

STUDENTS: DO NOT WRITE IN SHADED REGIONS

Answer		Scorer 2	Scorer 1
		0 or 1	0 or 1
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
5 th Grade		TOTAL:	

"Math Is Cool" Championships — 2025-26

5th Grade — Feb/March 2026

Final Score (out of 10)

Room #

School Name

Team #

Linda Moore Triple Jump - 15 minutes - ~15% of team score

This Submittal #2 will be collected after 10 minutes.

SUBMITTAL #2

STUDENTS: DO NOT WRITE IN SHADED REGIONS

Answer		Scorer 2 0 or 1	Scorer 1 0 or 1
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
5 th Grade		TOTAL:	

"Math Is Cool" Championships — 2025-26

5th Grade — Feb/March 2026

Final Score (out of 10)

Room #

School Name

Team #

Linda Moore Triple Jump - 15 minutes - ~15% of team score

This Submittal #3 will be collected after 15 minutes.

SUBMITTAL #3

STUDENTS: DO NOT WRITE IN SHADED REGIONS

Answer		Scorer 2 0 or 1	Scorer 1 0 or 1
1			
2			
3			
4			
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6			
7			
8			
9			
10			
5 th Grade		TOTAL:	

"Math Is Cool" Championships — 2025-26

5th Grade — Feb/March 2026

Room #

School Name

Team #

Total Score for Each Round

College Bowl #1 (10 Possible)	College Bowl #2 (10 Possible)	College Bowl #3 (10 Possible)

DO NOT USE TALLY MARKS ON THIS SHEET. WRITE THE TOTAL SCORE FOR EACH ROUND.

"Math Is Cool" Championships — 2025-26

5th Grade — Feb/March 2026

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"Math Is Cool" Championships — 2025-26

5th Grade — Feb/March 2026

Proctor
Copy

Mental Math Contest

MENTAL MATH - 30 seconds per question - ~25% of team score & ~8% of individual score

All students in the room will concurrently be asked the same eight questions in this individual test. When it is time to begin, the proctor will read the first question twice. You may not do any writing or talking while arriving at a solution. Once you have a solution, record it on the sheet in front of you. **You may not change or cross out answers once you have written an answer down. If there are eraser marks, write-overs, or crossed-out answers, they will be marked wrong.** Once all students have laid their pencils on the desk, another question will be asked. If a student doesn't lay his or her pencil down, the maximum wait time is 30 seconds after completion of the second reading of the question before the next question is read. You may continue to work on a problem (in your head) while the next question is being read. The raw score is 1 point per correct answer.

1	What is the sum of twenty-five and fifty-one?	
2	In inches, what is the perimeter of a square with a side length of six inches?	
3	What is fifty percent of sixty-eight?	
4	Charlie scores a fifty, forty-eight, fifty-five, sixty and sixty-two on his most recent math tests. Which score is the median?	
5	X plus fourteen equals thirty-four. What is x?	
6	A right triangle has a base of four feet and a height of five feet. A rectangle with a base of ten feet has the same area as the triangle. What is the height of the rectangle in feet?	
7	The letters of the alphabet A through Z are each assigned an integer from one to twenty-six in order. What is the total value of the letters in the word MATH, spelled M-A-T-H, when added together?	
8	The first two terms of a sequence are two and five. Each of the following terms in the sequence is equal to the sum of all the terms that come before it in the sequence. What is the eighth term in the sequence?	

"Math Is Cool" Championships — 2025-26

5th Grade — Feb/March 2026

Key

COLLEGE BOWL ROUND #1

#	Problem	Answer
1	One hundred twenty-three plus what number equals two hundred?	77
2	A train leaves at nine forty-five AM and arrives at its destination at one twenty PM on the same day. How many minutes did the trip take?	215 [minutes]
3	What is eight hundred fifty-two point four nine rounded to the nearest ten?	850
4	How many different ways are there to arrange the digits in the number two thousand twenty-six that do not have a zero in the first position?	9 [ways]
5	What is the minimum number of digits that must be removed from the number twelve million three hundred twenty-three thousand three hundred fourteen so that the remaining digits are a palindrome?	3 [digits]
6	Leo is going to fill a box with measurements forty by forty by sixty units, using identical cubes of some size. What is the minimum number of cubes needed?	12 [cubes]
7	A movie theater has twelve rows of seats with eighteen seats in each row. One show sold one hundred sixty-eight tickets. How many empty seats were there?	48 [seats]
8	What is the next number in the pattern: Five, ten, twenty, forty, and so on	80
9	Mia's water bottle can hold twenty-four ounces of water. She drinks two full water bottles each day. How many total cups of water does Mia drink each day?	6 [cups]
10	A book has two hundred fifty-six pages. If a student reads thirty-two pages each day, how many full days will it take to finish the book?	8 [days]

"Math Is Cool" Championships — 2025-26

5th Grade — Feb/March 2026

Key

COLLEGE BOWL ROUND #2

#	Problem	Answer
1	What is the smallest 3-digit integer sum of two prime numbers?	100
2	What is the next number in the following sequence? Seventeen, eleven, five, negative one, and so on.	-7
3	Milo, Nim, Tyler and Lacy took turns in that order saying the whole numbers from one to twenty-five. How many times did Milo say a number that contains the digit one?	4 [times]
4	The length of a rectangle is eight inches, and the width is half as long. In inches, what is the side length of a square with the same perimeter as the rectangle?	6 [in]
5	X plus X plus six equals X plus X plus X plus X. What is the value of X plus two?	5 [= $x + 2$]
6	The sum of the digits of a six-digit whole number is five. What is the product of the digits?	0 [= product]
7	Julian had fifty dollars saved. He bought two board games that each cost eighteen dollars. Then he earned twelve dollars by doing chores. How many dollars does Julian have now?	26 [dollars]
8	In a line of students, there are twenty-seven students in front of Ivy and thirteen students behind her. Stefan is exactly in the middle. How many students are between Ivy and Stefan?	6 [students]
9	A rectangle is sixteen inches long and three inches wide. In square inches, what is one-half of its area?	24 [square inches]
10	A pizza was cut into twelve equal slices. Three friends each ate three slices. What percent of the pizza is left?	25 [%]

"Math Is Cool" Championships — 2025-26

5th Grade — Feb/March 2026

Key

COLLEGE BOWL ROUND #3

#	Problem	Answer
1	Rachel buys three notebooks for four dollars each and two pens for two dollars each. She pays with a twenty dollar bill. How many dollars does she get back?	4 [dollars]
2	The area of a rectangle is forty-eight square units. If the width is six units, what is the length in units, times two?	16 [units]
3	A ten-sided die is numbered zero through nine. On a single roll, what is the probability in percent of rolling a number less than four?	40 [%]
4	The first three terms in an arithmetic sequence are: twenty-one halves, eleven, and twenty-three halves. What is the 6 th term?	13 [= 6 th term]
5	Using the digits one through eight exactly once each, form two four-digits numbers with a sum as small as possible. What is the sum?	3825 [= sum]
6	A standard deck of playing cards has the Jacks, Queens and Kings removed from the deck. If one card is randomly selected, what is the probability in percent that it is an ace or a two?	20 [%]
7	A group of students spend the day reading. Three students read for twenty minutes, four for thirty minutes, and three for forty minutes. How many hours did the students spend reading in total?	5 [hours]
8	Solve for X: Four X plus twelve equals forty-four	8 [=x]
9	How many total sides do three octagons, two hexagons and one trapezoid have?	40 [sides]
10	A pizza costs fifteen dollars. How many can you buy with ninety dollars?	6 [pizzas]

"Math Is Cool" Championships — 2025-26

5th Grade — Feb/March 2026

Key

COLLEGE BOWL ROUND #4

#	Problem	Answer
1	A recipe for chocolate chip cookies calls for two-thirds cup of sugar. How many cups of sugar is needed if Jenny wants to triple the recipe?	2 [cups]
2	Solve for X: Twenty minus X equals seven minus eight	21 [= x]
3	Connor scored eight hundred two thousand one hundred fourteen points playing an arcade game. Esmat scored seven hundred ninety-eight thousand one hundred ten points in the game. How many more points did Connor score?	4,004 [points]
4	The mean of the following numbers is twenty-four. What is the value of X? X, twelve, fourteen, forty	30 [= x]
5	Jose built a square outer wall using thirty-six identical unit cubes. How many unit cubes are needed to completely fill the interior of the square?	64 [cubes]
6	How many whole numbers X are there for which X minus twenty-four and X plus twenty-four are two-digit numbers?	42 [whole numbers]
7	A sequence begins with: One, negative one, negative one, one, negative one, and so on. Each number from the third on is equal to the product of the two preceding numbers. What is the sum of the first ten numbers?	-2 [= sum]
8	Rafa rolled a six-sided die four times, and scored a total of twenty-three points. How many sixes did he roll?	3 [sixes]
9	Sophie and her mom want to rent a canoe. Canoe rentals cost forty-five dollars for a half hour. If Sophie and her mom rent a canoe for one hour and thirty minutes, how many dollars will it cost?	135 [dollars]
10	What is two-thirds of two hundred seventy?	180

"Math Is Cool" Championships — 2025-26

5th Grade — Feb/March 2026

Key

COLLEGE BOWL ROUND #5

#	Problem	Answer
1	What is one hundred forty-four times three-fourths?	108
2	The side lengths of a triangle are six, ten and eleven units. An equilateral triangle has the same perimeter as this triangle. How long is one side of the equilateral triangle, in units?	9 [units]
3	The third through fifth terms of an arithmetic sequence are one-half, three-fourths, and one. What is the first term?	0 [= 1 st term]
4	How many of the whole numbers from one through nine is the following number divisible by? Two million three hundred ninety-two thousand six hundred forty-three	1 [number]
5	Ian has a stapler, a ruler, a calculator and a compass. How many different ways can he line them up on his desk if the calculator and compass both have to be next to the stapler?	4 [ways]
6	What is the units digit when the first five odd whole numbers are multiplied together?	5 [= units digit]
7	Maria has forty-nine blue pencils and one red pencil. How many blue pencils does she have to get rid of so that ninety percent of her pencils are blue?	40 [pencils]
8	Packard is twenty-one years old. In years, what is the mean of his age seven years ago and his age eleven years ago?	12 [years]
9	Two million equals two hundred times what number?	10,000 [ten thousand]
10	Ten runners are in a cross-country race. At the finish, there are three more runners behind Oliver than there are in front of him. In what position number did Oliver finish?	4 [or 4 th]

"Math Is Cool" Championships — 2025-26

5th Grade — Feb/March 2026

Key

COLLEGE BOWL ROUND #6

#	Problem	Answer
1	A crate is 6 feet long, 24 inches wide, and 3 feet tall. What is the volume in cubic feet?	36 [feet cubed]
2	Find the value of x that makes the expression true. X plus fifteen equals thirty plus two	17
3	What is the product of seven, three and five?	105 [= product]
4	June has some red and some black marbles in a bag, with a total of twenty-five marbles. If one marble is randomly selected, the probability that it is red equals X and the probability that it is black equals four X . How many black marbles are there?	20 [black marbles]
5	How many times more even divisors than odd divisors does the number thirty-six have?	2 [times]
6	A bicyclist travels five meters in one second. The wheels on the bike have a circumference of one hundred twenty-five centimeters. How many complete revolutions does each wheel make in five seconds?	20 [revolutions]
7	Alex has a one-meter-long rope and a two-meter-long rope. He cuts up both ropes so that the resulting pieces are all of equal length. What is the greatest number of pieces less than twenty that he can obtain in this manner?	18 [pieces]
8	A sequence begins with zero, five, ten, fifteen, and so on. What is the sum of the first eight numbers in the sequence?	140 [= sum]
9	How many inches are in nine point two five feet?	111 [inches]
10	What is the perimeter of a square divided by the sum of two side lengths of the square?	2

"Math Is Cool" Championships — 2025-26

5th Grade — Feb/March 2026

Key

COLLEGE BOWL - EXTRA QUESTIONS

#	Problem	Answer
1	Jasmine has forty-one cents in coins, where each coin has a different value. How many coins does she have?	4 [coins]
2	How many equilateral triangles with side length five units can fit inside a regular hexagon with side length five units?	6 [equi. Triangles]
3	A bucket holds eighty cups of water. How many gallons of water does the bucket hold?	5 [gallons]
4	Biff bought a box of candies on Tuesday, and ate half of them right away. He ate ten more on Wednesday. On Thursday there were only five left. How many candies were in the box?	30 [candies]
5	It takes Josh two and a half hours to hike around a mountain. How many minutes does it take him?	150 [minutes]
6	What is the maximum number of points of intersection between a triangle and a circle?	6 [points]
7	A movie theater has twelve rows of seats. Each row has sixteen seats. There are also ten extra seats in the front. How many seats are in the theater?	202 [seats]
8	Two integers have a sum of sixteen and a difference of two. What is their product?	63 [= product]
9	What is the next number in the sequence that begins with: One, three, seven, thirteen, twenty-one, and so on?	31 [= next number]
10	Carlos got seventeen questions correct out of twenty on a quiz. What percent of the questions did he get wrong?	15 [%]